

The Central Dogma of a Cell & DNA Replication

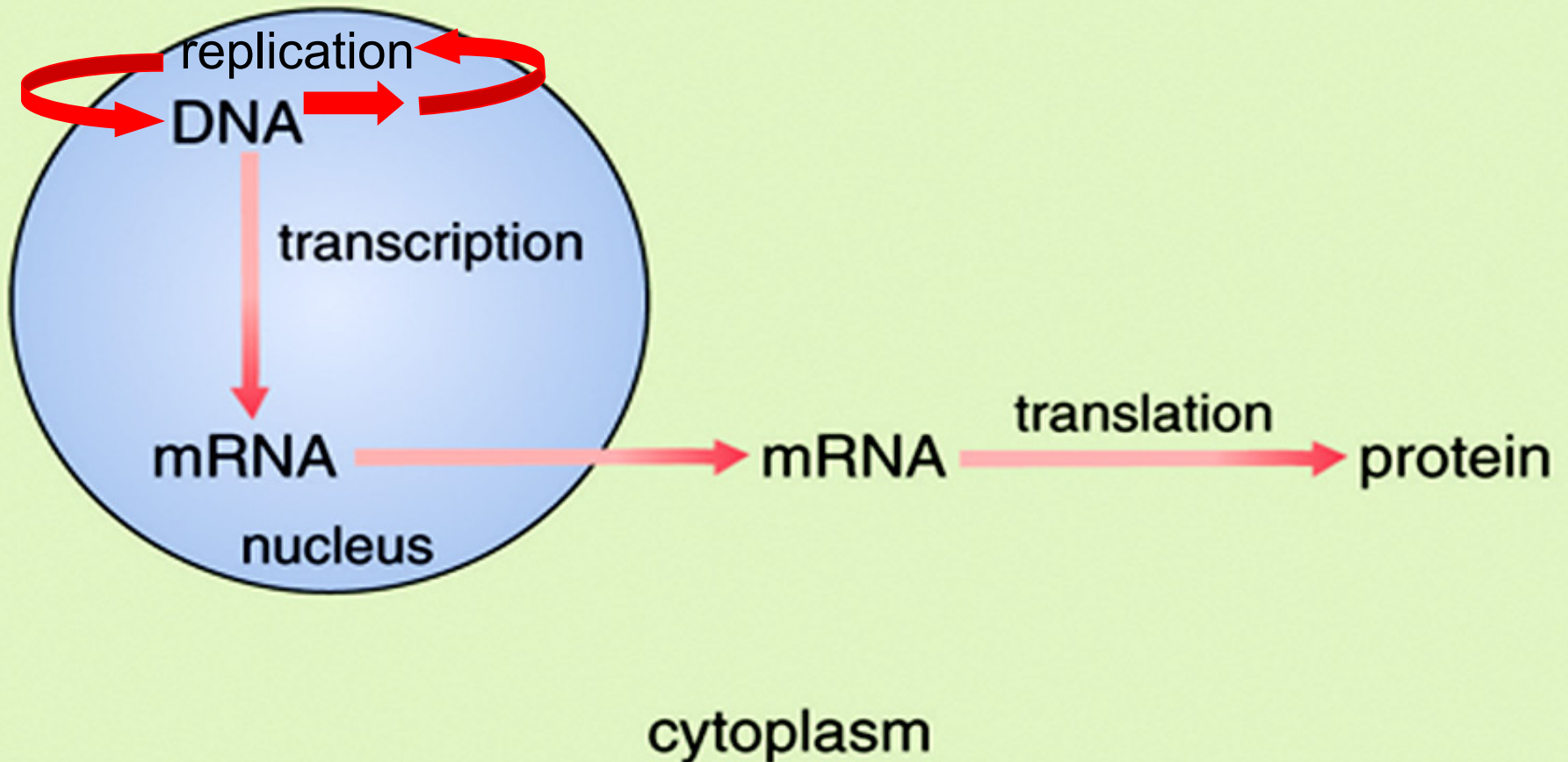
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Objectives:

At the end of this chapter, it is expected;

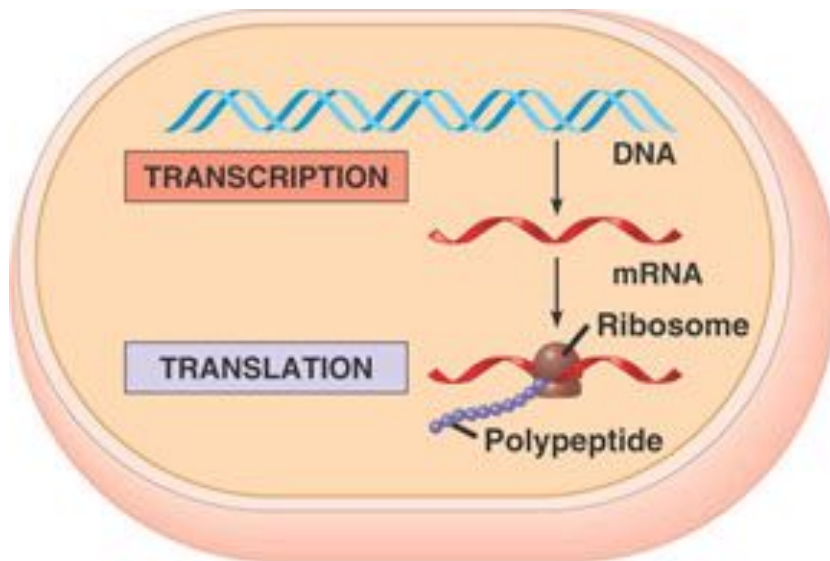
- To explain the central dogma of the cell.
- To describe the models of DNA replication
- To appreciate the features of DNA replication
- To describe the roles of enzymes and proteins involve in DNA Replication
- To describe the stages and steps of DNA replication
- To explain telomeres problem in eukaryotic DNA replication
- To differentiate between replication in prokaryotes and eukaryotes
- To name the inhibitors of DNA replication and explain their mechanism

The Central Dogma of a Cell

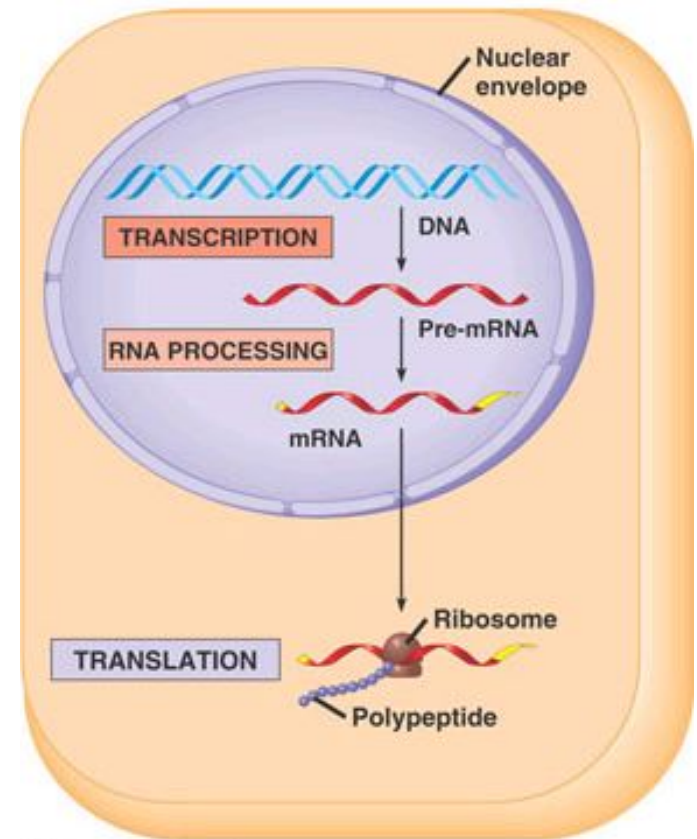


The Central Dogma

DNA replicated into *DNA* transcribed into *mRNA*
translated into *protein*



(a) Prokaryotic cell

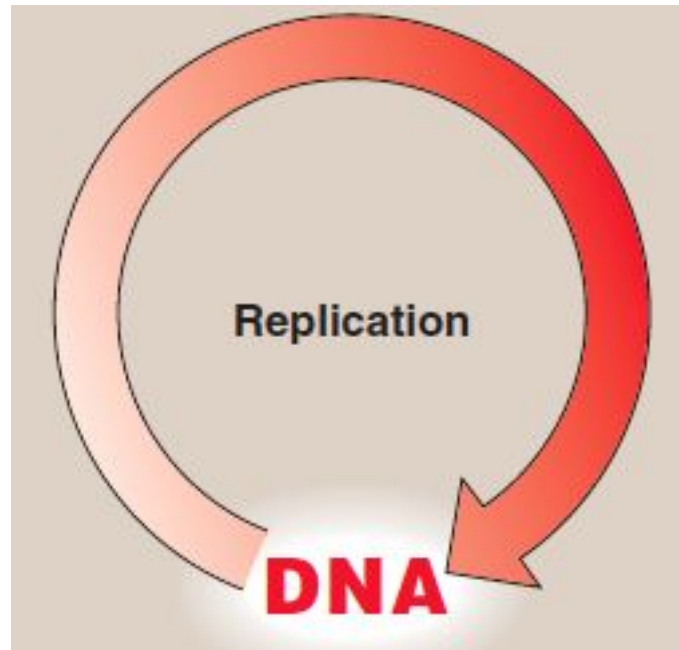


(b) Eukaryotic cell

DNA Replication

DNA Replication

Process by which DNA makes a copy of itself during cell division



Models of DNA Replication

- Proposed by Matthew Meselson & Franklin Stahl (1958).
- 3 possible mechanisms:
 - ❖ **Conservative model**
 - ❖ **Semiconservative model**
 - ❖ **Dispersive model**

Models of DNA Replication

a. **Conservative model:**

- Both strands of one copy would be entirely old DNA, while the other copy would have both strands of new DNA.

b. **Dispersive model:**

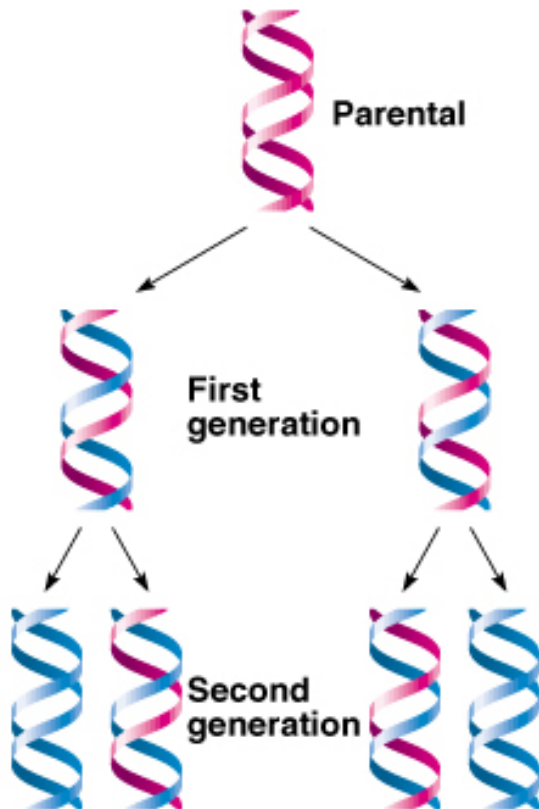
- dsDNA might fragment, replicate and then reassemble, creating a mosaic of old and new dsDNA regions in each new chromosome.

c. **Semiconservative model:**

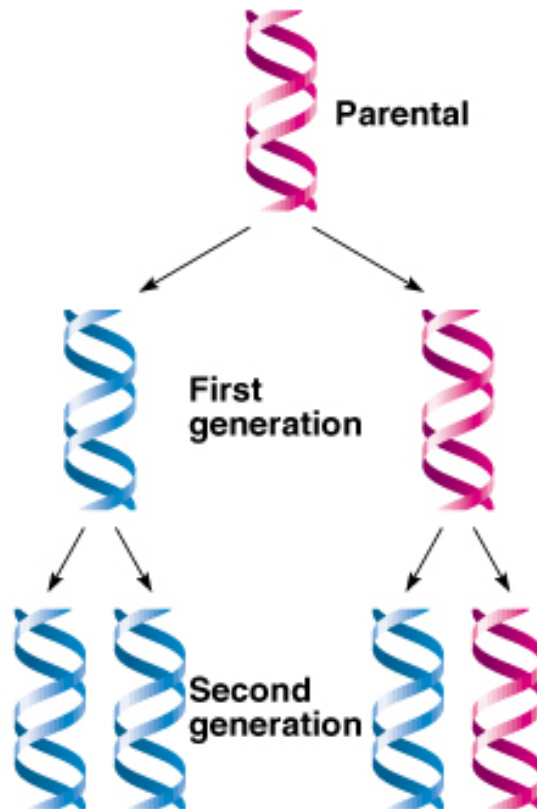
- DNA strands separate, and a complementary strand is synthesized for each.
- Sibling chromatids have one old and one new strand.
- This model was the winner in the Meselson and Stahl experiment.

Three models for the replication of DNA

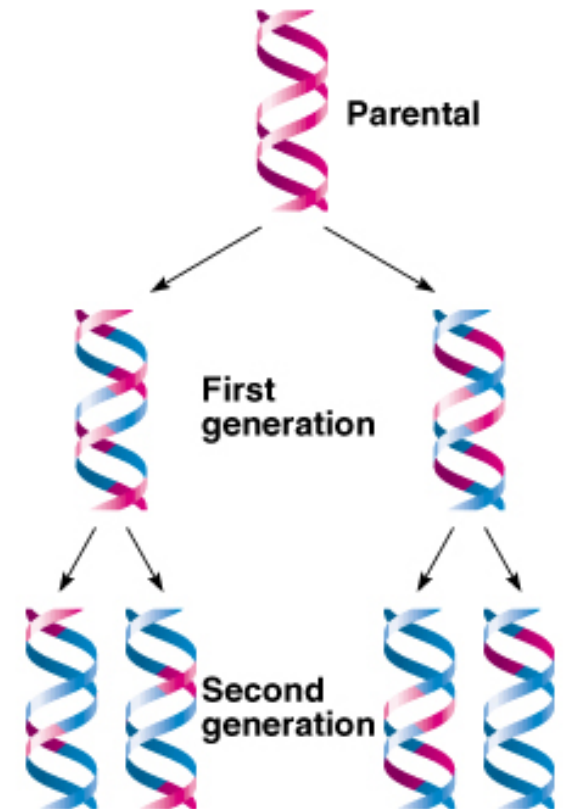
a) The semiconservative model



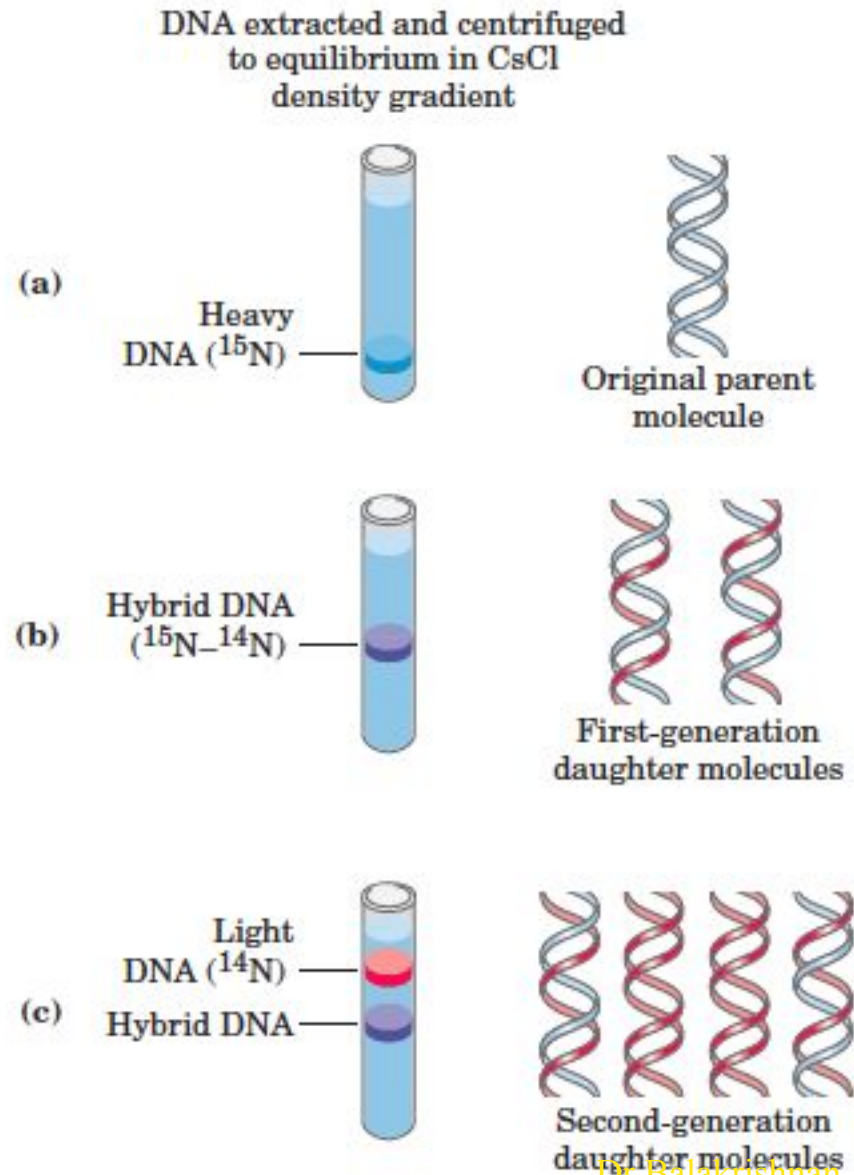
b) The conservative model



c) The dispersive model



The Meselson-Stahl experiment



SALIENT FEATURES OF DNA REPLICATION

- Provision of progeny with the genetic make up possessed by the parents - primary function.
- DNA replication must be complete, to be carried out in such a way as to maintain genetic stability within the organism and the species.
- To ensure fidelity in replications, it involves many cellular functions through a number of protein-protein and protein-DNA interactions:
 - Nucleotide selectivity of DNA Pol
 - Exonucleolytic proofreading procedures, and
 - Postreplicative DNA mismatch repair
- In all cells, replication can occur only from a ssDNA template, which occurs after unwinding dsDNA.
- Mechanisms must exist to target the site of initiation of replication, where the replication complex is formed.
- After replication is complete in an area, the parent and daughter strand must reform dsDNA.
- In Eukaryotic cells, an additional step is must, in that domain structure must reform, including nucleosomes, that existed prior to the onset of replication.

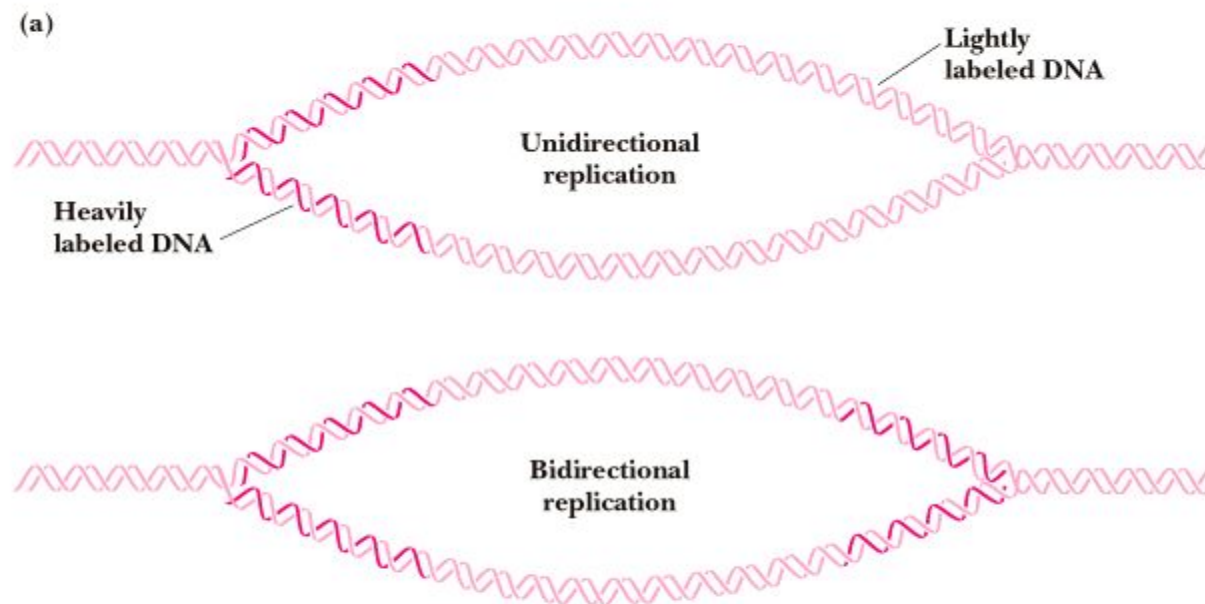
Features of DNA Replication:

DNA replication is **bidirectional**;

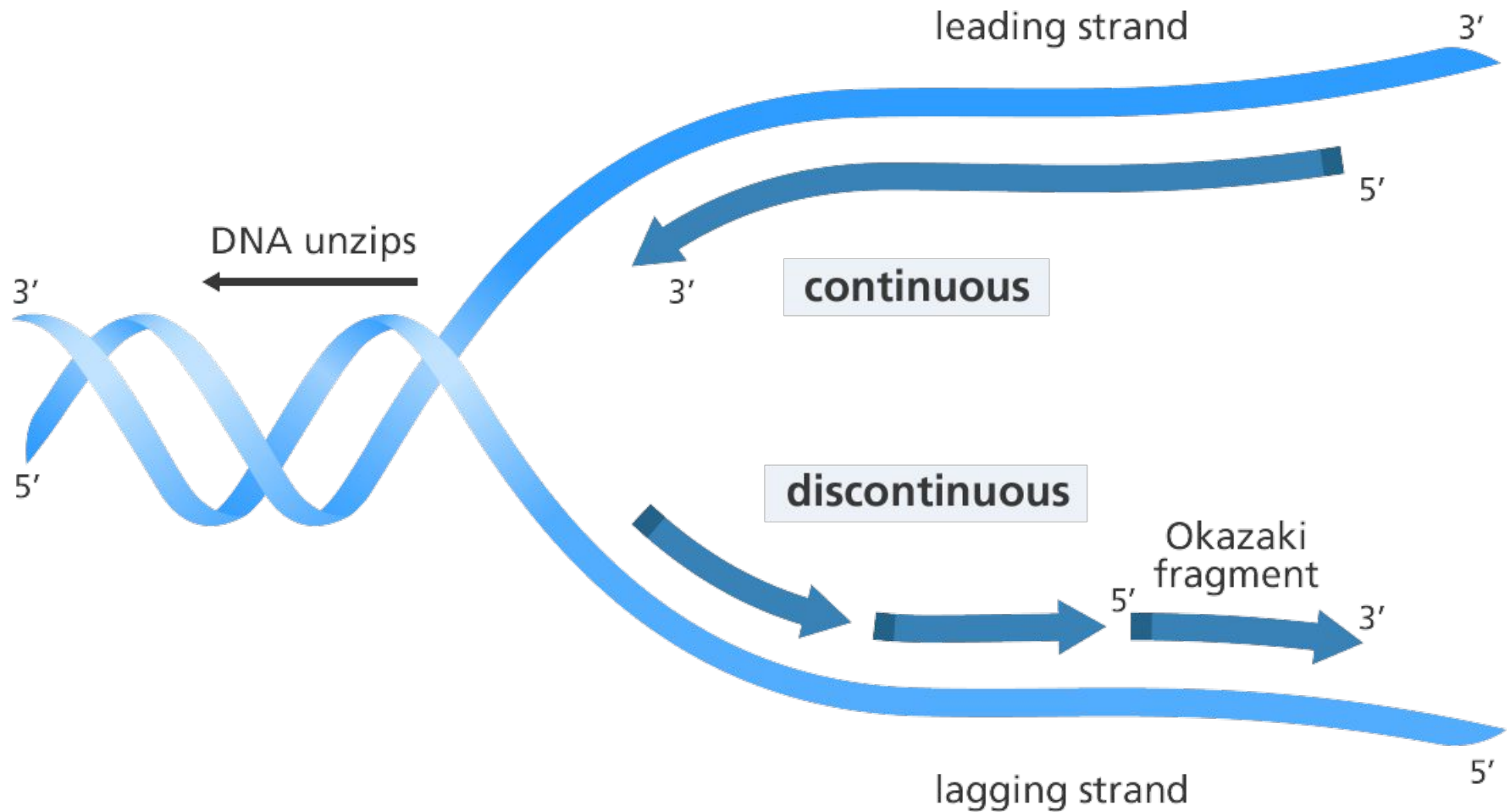
- Bidirectional replication involves two replication forks, which move in opposite directions

DNA replication is **semidiscontinuous**;

- The leading strand copies continuously
- The lagging strand copies in segments (**Okazaki fragments**) which must be joined



DNA replication fork



An illustration to show replication of the leading and lagging strands of DNA.

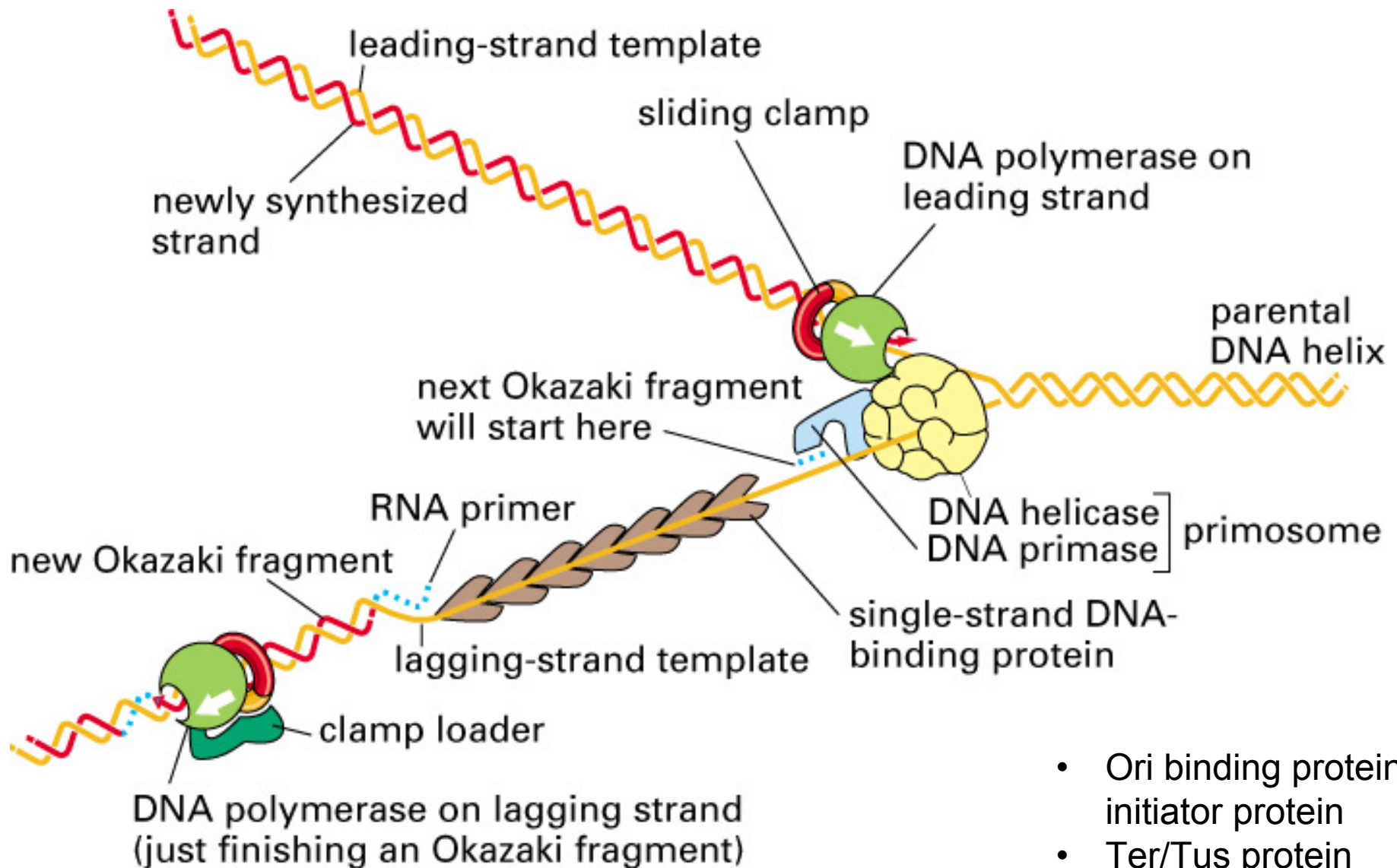
Most important Enzymes involve in DNA Replication

1. **DNA helicase:**
 - Unwinds the DNA double helix.
2. **DNA gyrase (topoisomerase II):**
 - Keep the double strands from tangling during replication (release the Torsional stress)
 - Helps unwind the DNA double helix
3. **RNA primase:**
 - Builds an RNA primer on the parent strand to initiate DNA replication
4. **DNA polymerases (Pol I, II, & III/ Pol α , β , γ , δ & ϵ):**
 - Matches and lays down nucleotides to build the daughter DNA strand along each parent DNA strand
5. **DNA ligase:**
 - Joins the adjacent Okazaki fragments on the lagging strand of DNA
6. **Telomerase:**
 - Involves in synthesis of telomeres of eukaryotic DNA

A comparison of prokaryotic and eukaryotic DNA polymerases.

<i>E coli</i>	Mammalian	Function
I	α	Gap filling and synthesis of lagging strand
II	ϵ	DNA proofreading and repair
	β	DNA repair
	γ	Mitochondrial DNA synthesis
III	δ	Processive, leading strand synthesis

Protein machinery for DNA replication



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Figure 5-21. Molecular Biology of the Cell, 4th Edition.

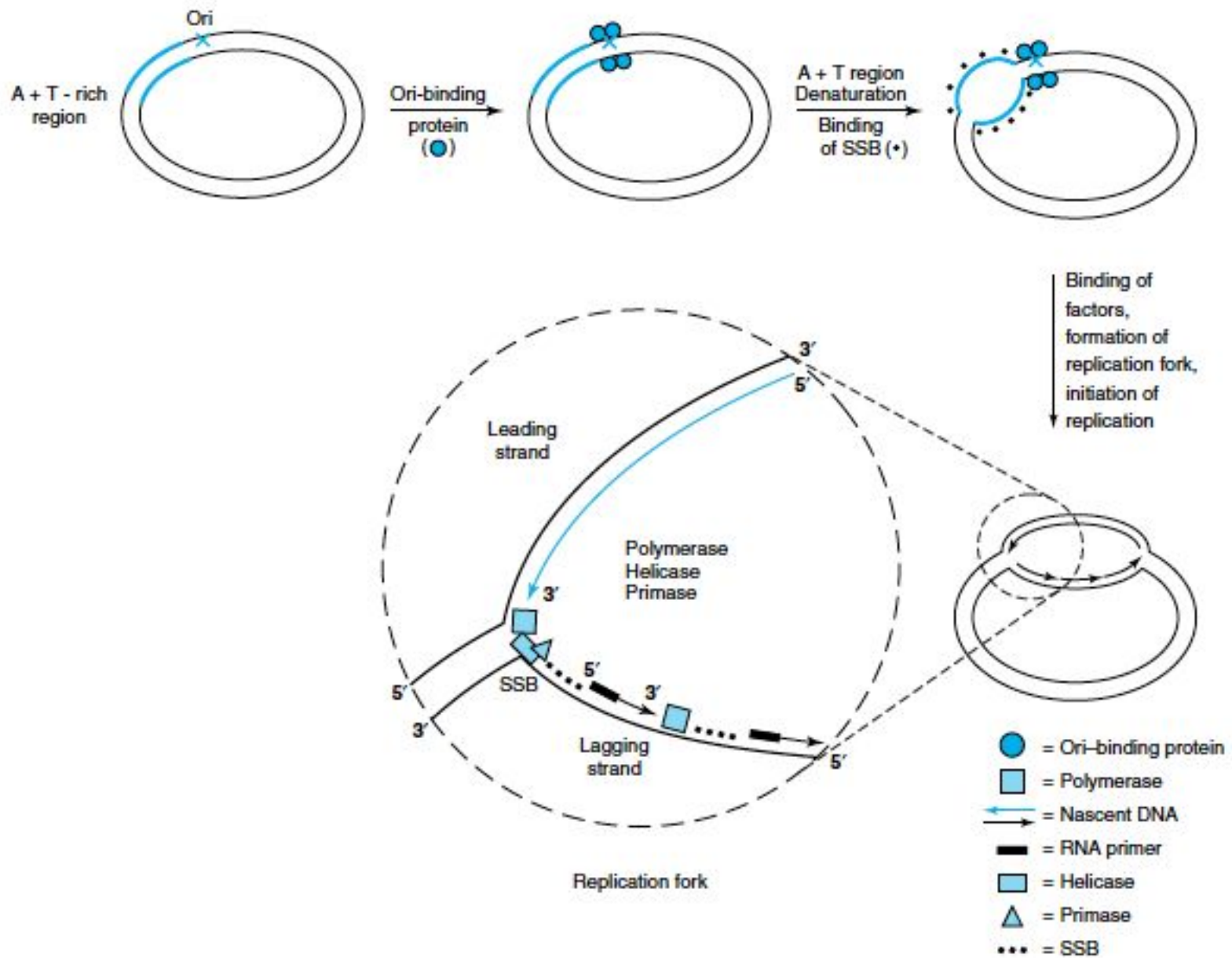
Stages of DNA Replication

DNA replication includes:

1. Initiation
2. Elongation
3. Termination

2. Initiation:

- Begins at an **origin of replication**
- Ori binding protein (e.g. DnaA protein in *E coli*) binds to repeats in ori, initiating strand separation at A+T rich region
- DnaB, a helicase delivered by DnaC, further unwinds.
- Primase then binds and constructs the RNA primer. 17

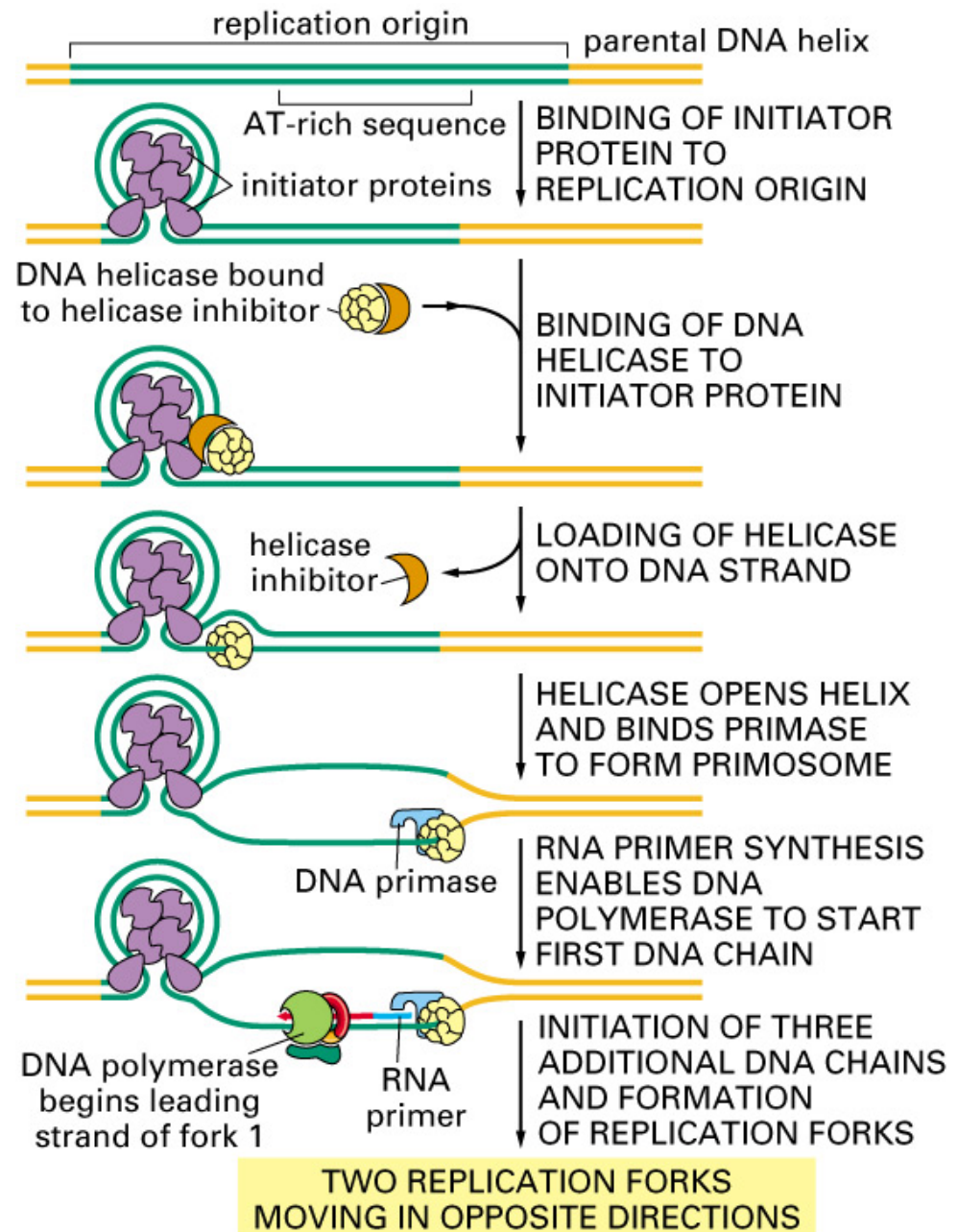


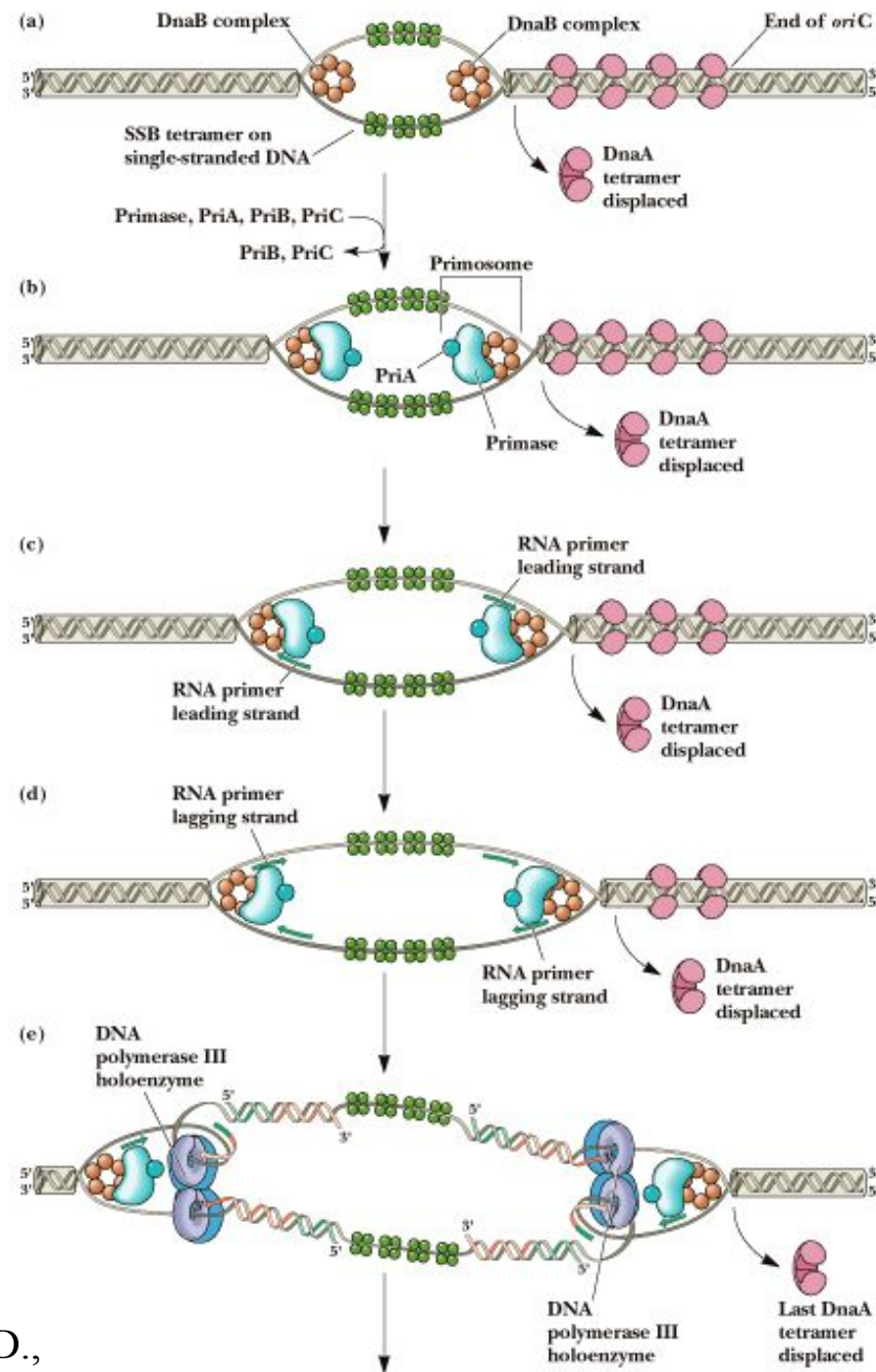
Steps of DNA Replication:

1. The double helix structure of the DNA molecule ‘unzips’ by helicase
 - Helicase breaks the hydrogen bonds holding the complementary bases of DNA together (A with T, C with G).
2. The separation of the two single strands of DNA creates a ‘Y’ shape called a replication ‘fork’.
 - The two separated strands will act as templates for making the new strands of DNA.
 - Supercoiling must be compensated by DNA gyrase
 - One of the strands is oriented in the 3’ to 5’ direction (towards the replication fork), this is the leading strand.
 - The other strand is oriented in the 5’ to 3’ direction (away from the replication fork), this is the lagging strand.
 - As a result of their different orientations, the two strands are replicated differently (shown in the following slides).

Initiating for DNA replication

1. Initiator protein,
2. Helicase binding to initiator protein,
3. Helicase loading on DNA,
4. Helicase opens the DNA and binds to primase,
5. RNA primer synthesis,
6. DNA polymerase binding and DNA synthesis





2. Elongation:

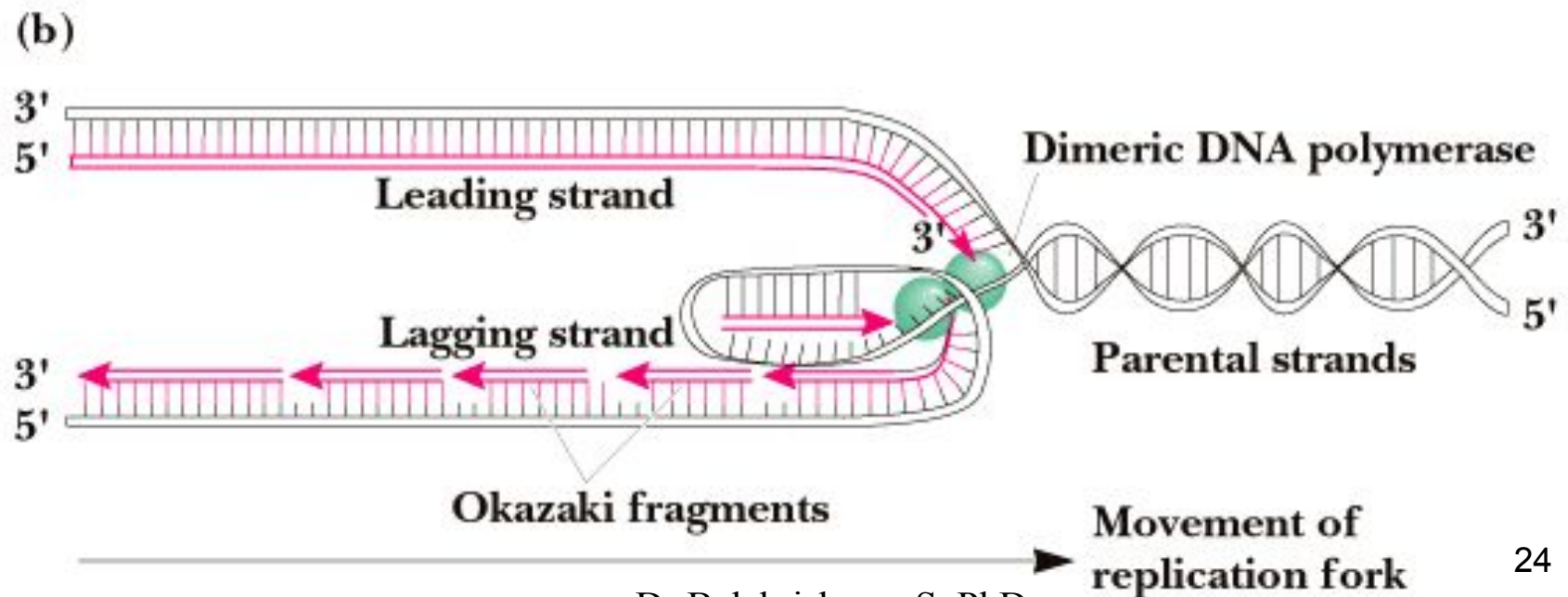
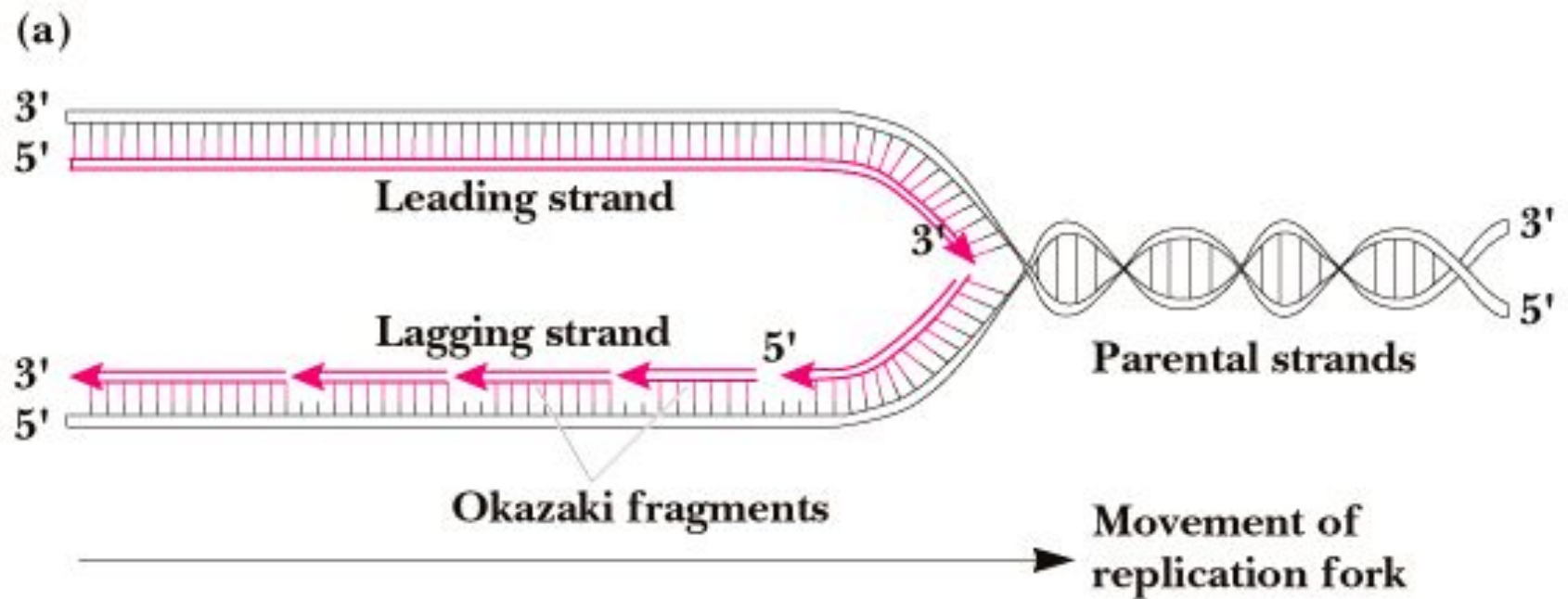
- New strands of DNA are synthesized by **DNA polymerase**
- Involves:
 - DnaB helicase unwinding,
 - SSB binding to keep strands separated, and
 - DNA polymerase grinding away on both strands

Leading Strand:

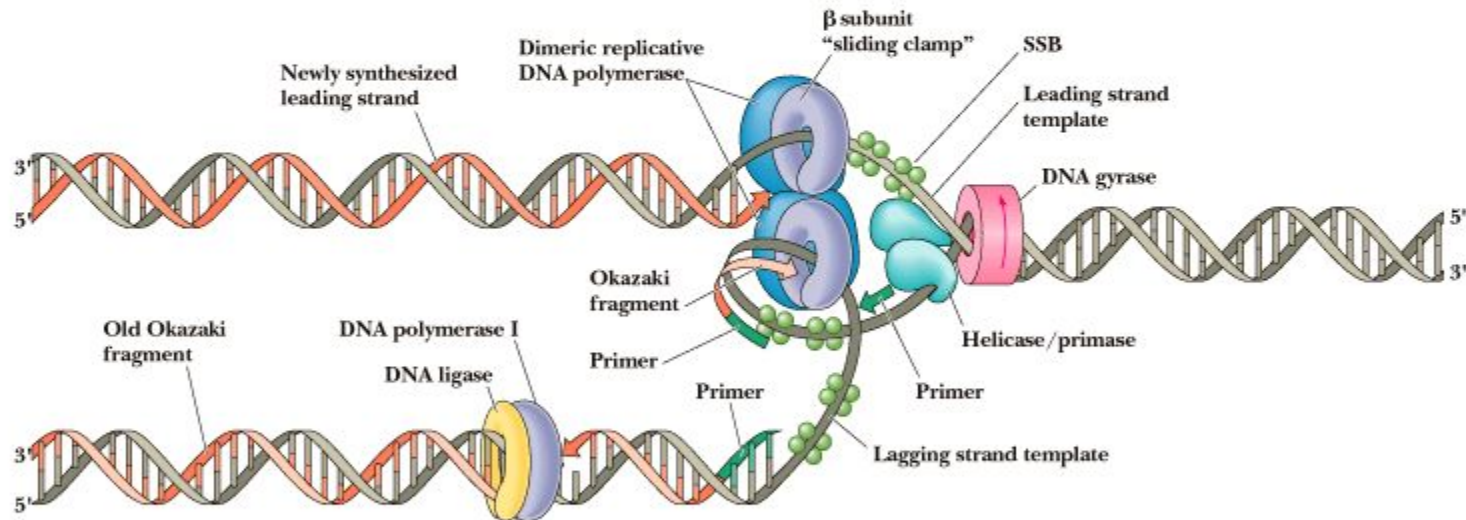
3. A primer comes along and binds to the end of the leading strand. The primer acts as the starting point for DNA synthesis.
4. DNA polymerase (pol III; pol δ) binds to the leading strand and then 'walks' along it, adding new complementary nucleotide bases to the strand of DNA in the 5' to 3' direction.
5. This sort of replication is called continuous.

Lagging strand:

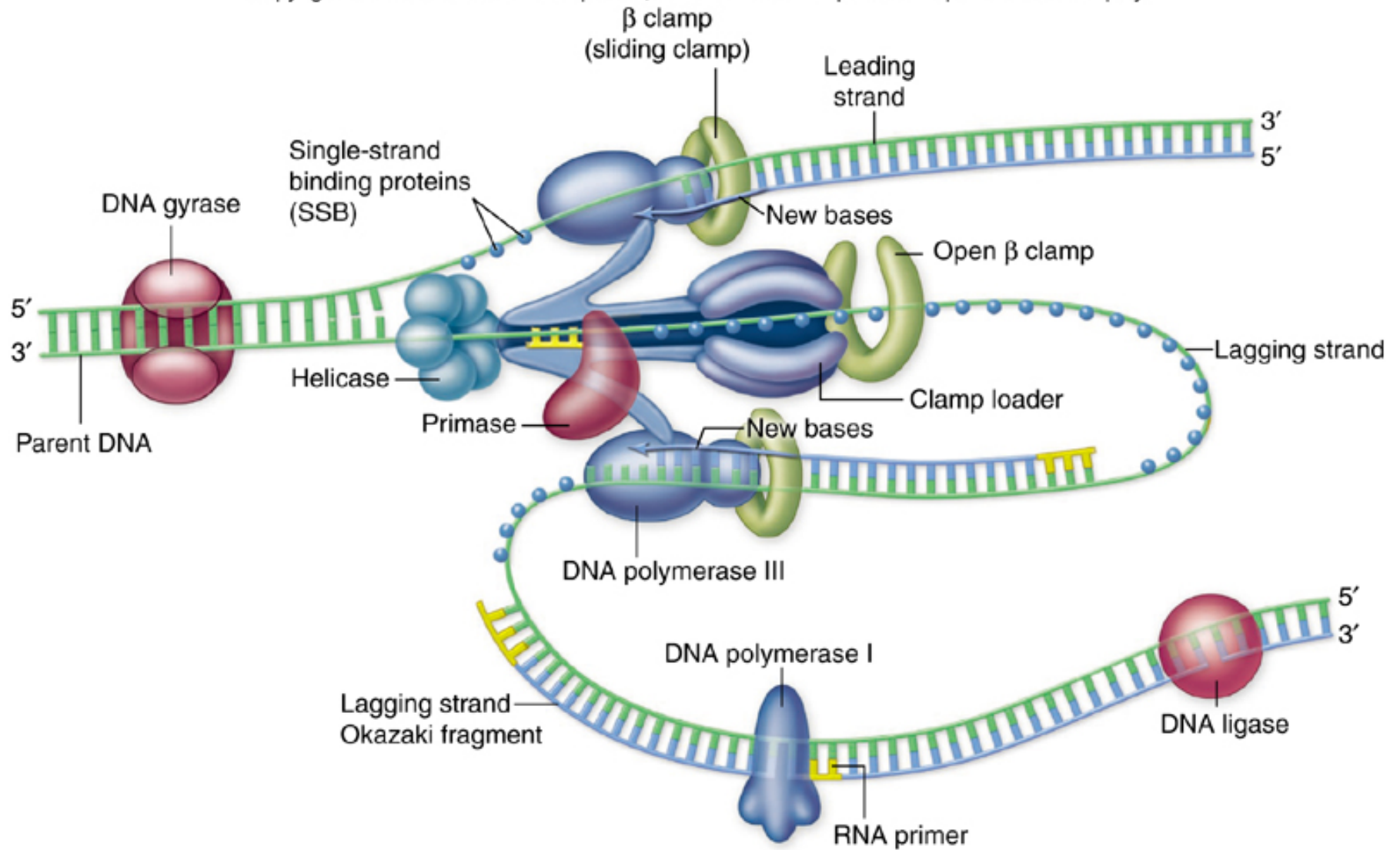
3. Numerous RNA primers are made by the primase enzyme and bind at various points along the lagging strand.
4. Chunks of DNA called Okazaki fragments are then added to the lagging strand also in the 5' to 3' direction. (pol I; pol α)
5. This type of replication is called discontinuous as the Okazaki fragments will need to be joined up later.



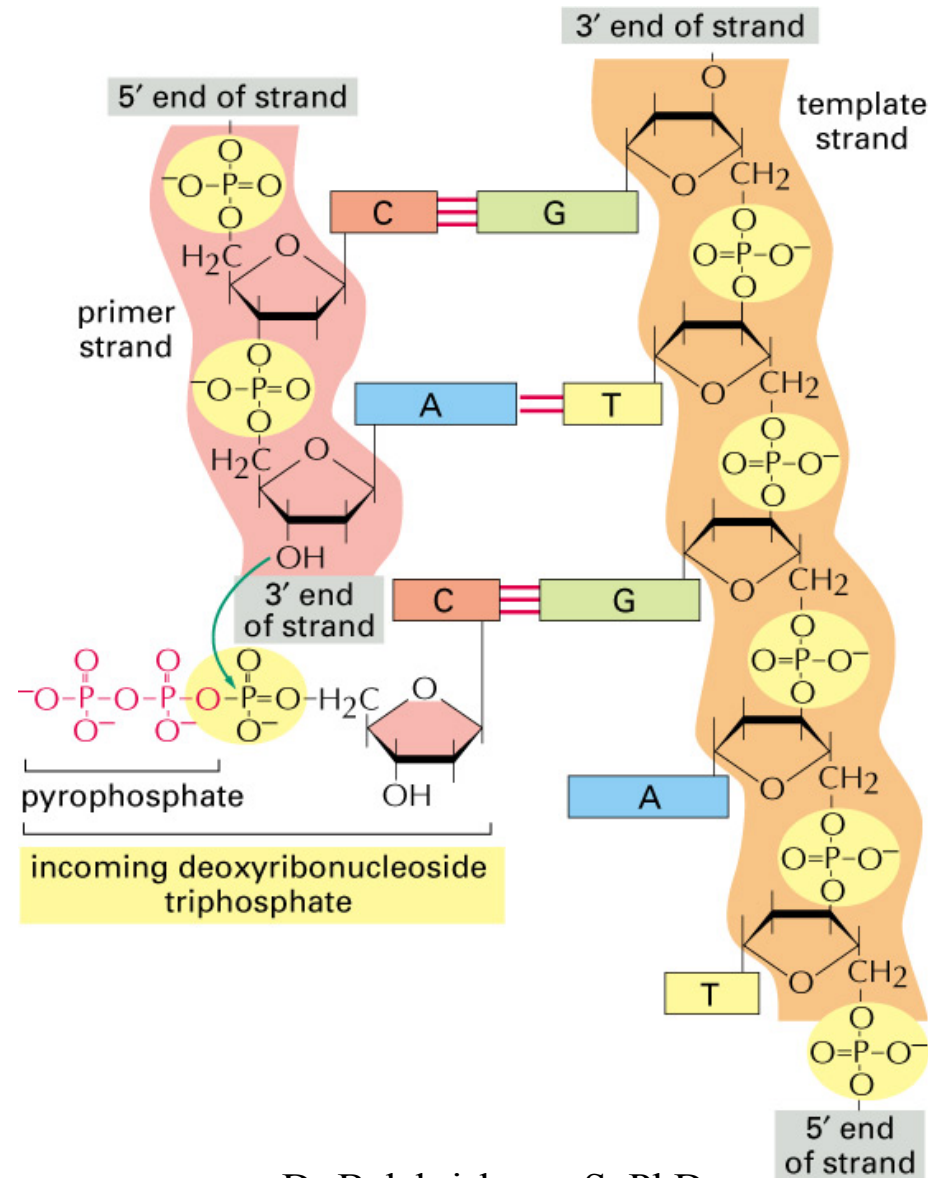
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Figure 30.15



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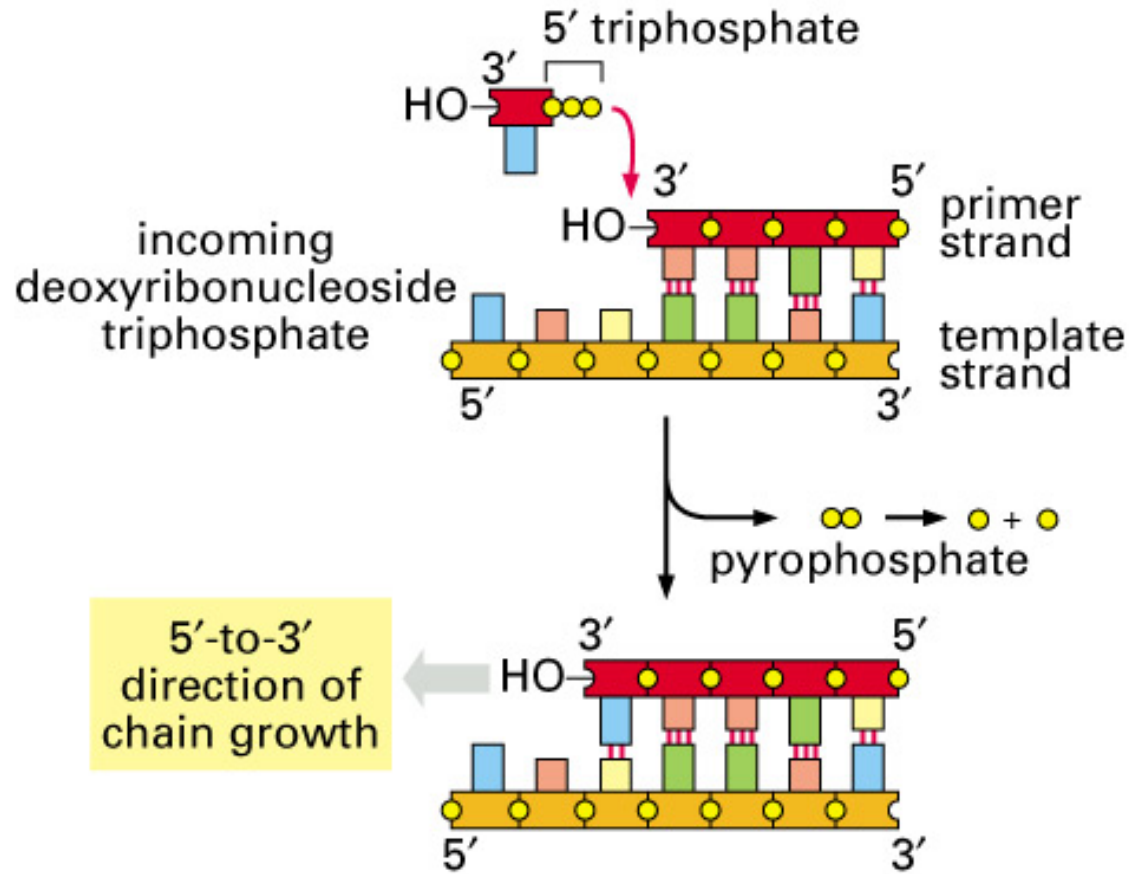
The Chemistry of DNA replication



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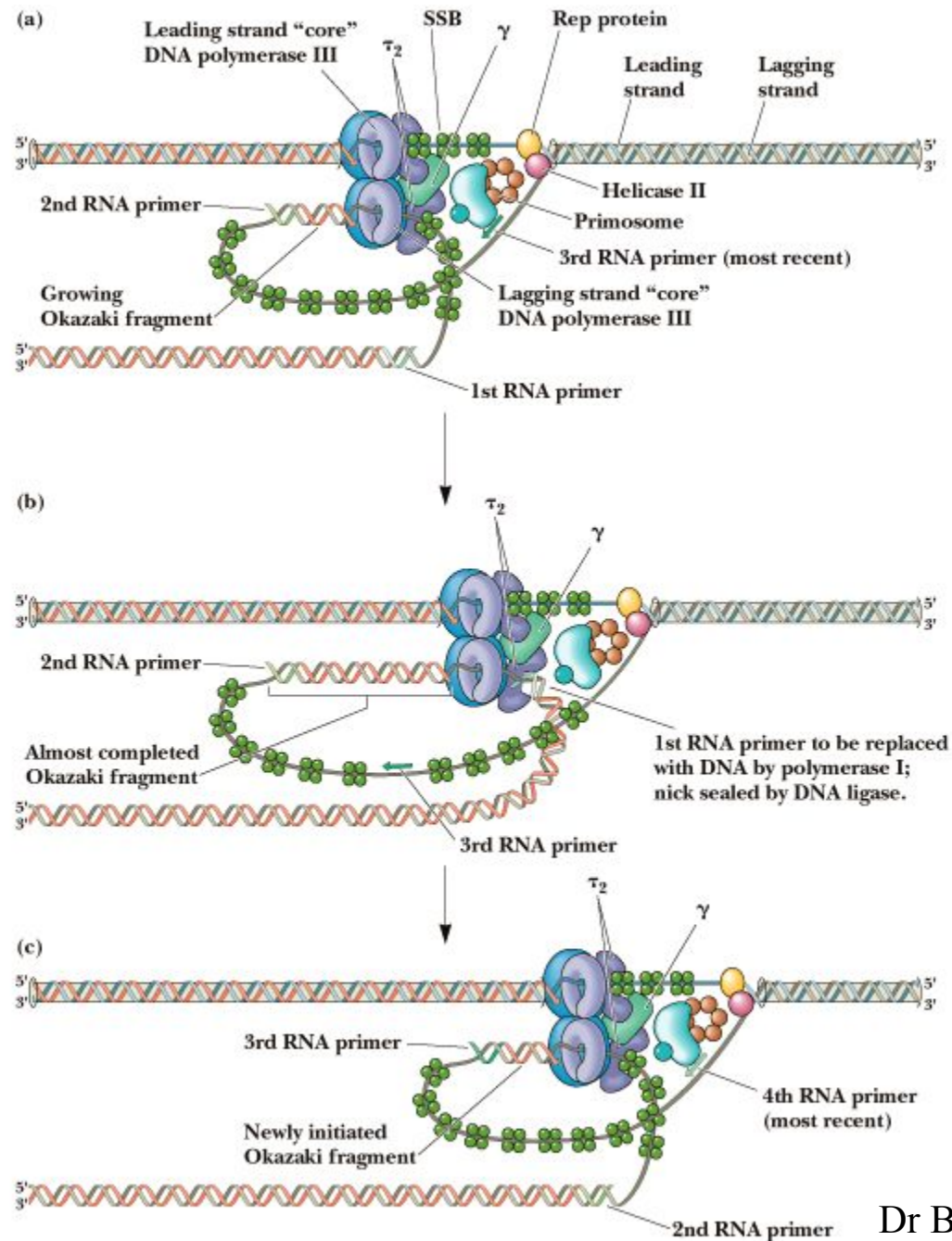
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DNA Synthesis by DNA polymerase

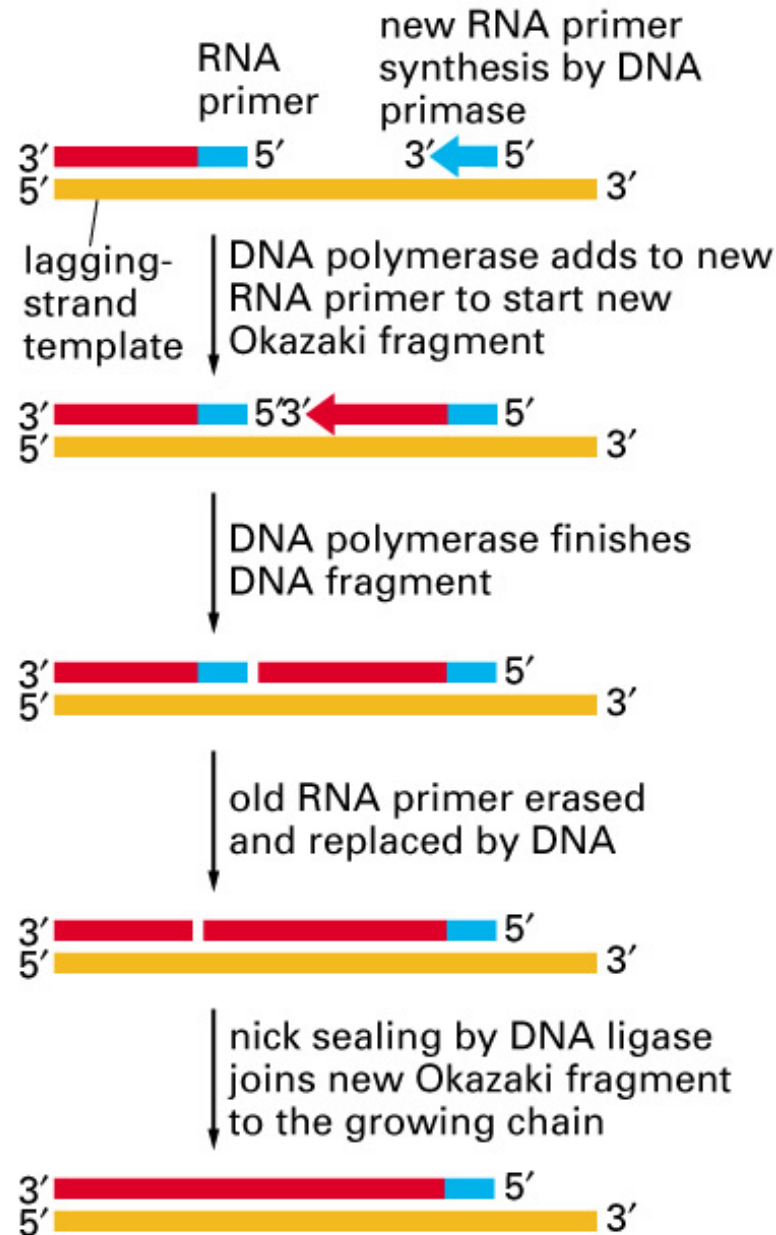


(A)

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DNA Replication at the Lagging strand



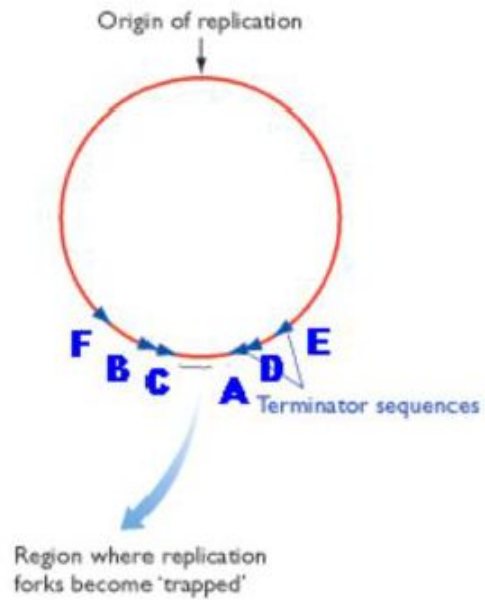
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Figure 5–13. Molecular Biology of the Cell, 4th Edition.

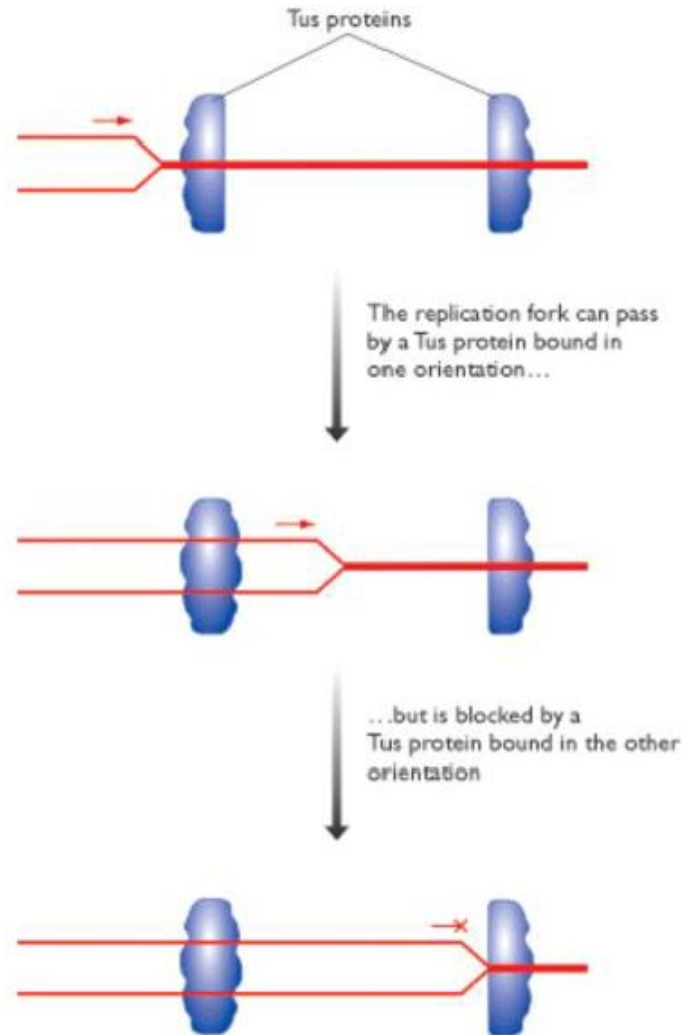
3. Termination:

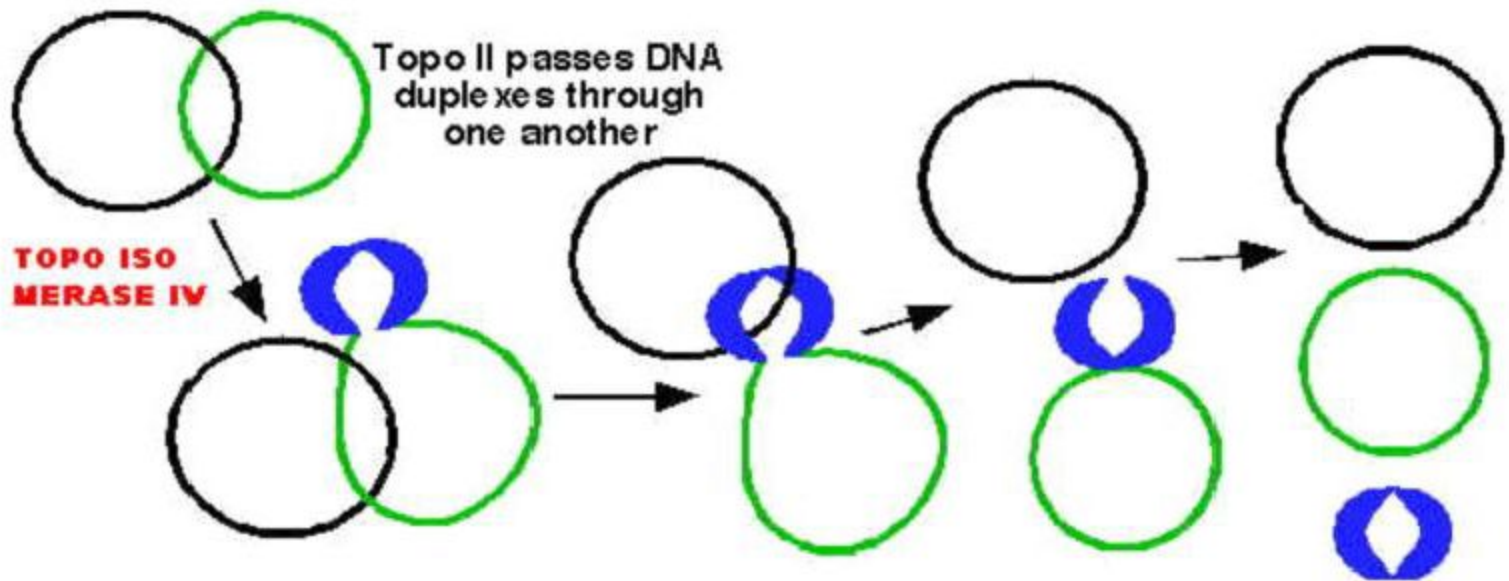
- Terminated differently in prokaryotes and eukaryotes
- The "ter" locus, rich in Gs and Ts, signals the end of replication. A Ter/Tus protein is also involved.
- Ter/Tus protein is a contrahelicase and prevents unwinding
- In eukaryotes - when replisomes from adjacent replication origins converge

(A) Terminator sequences in the *E. coli* genome



(B) The role of Tus

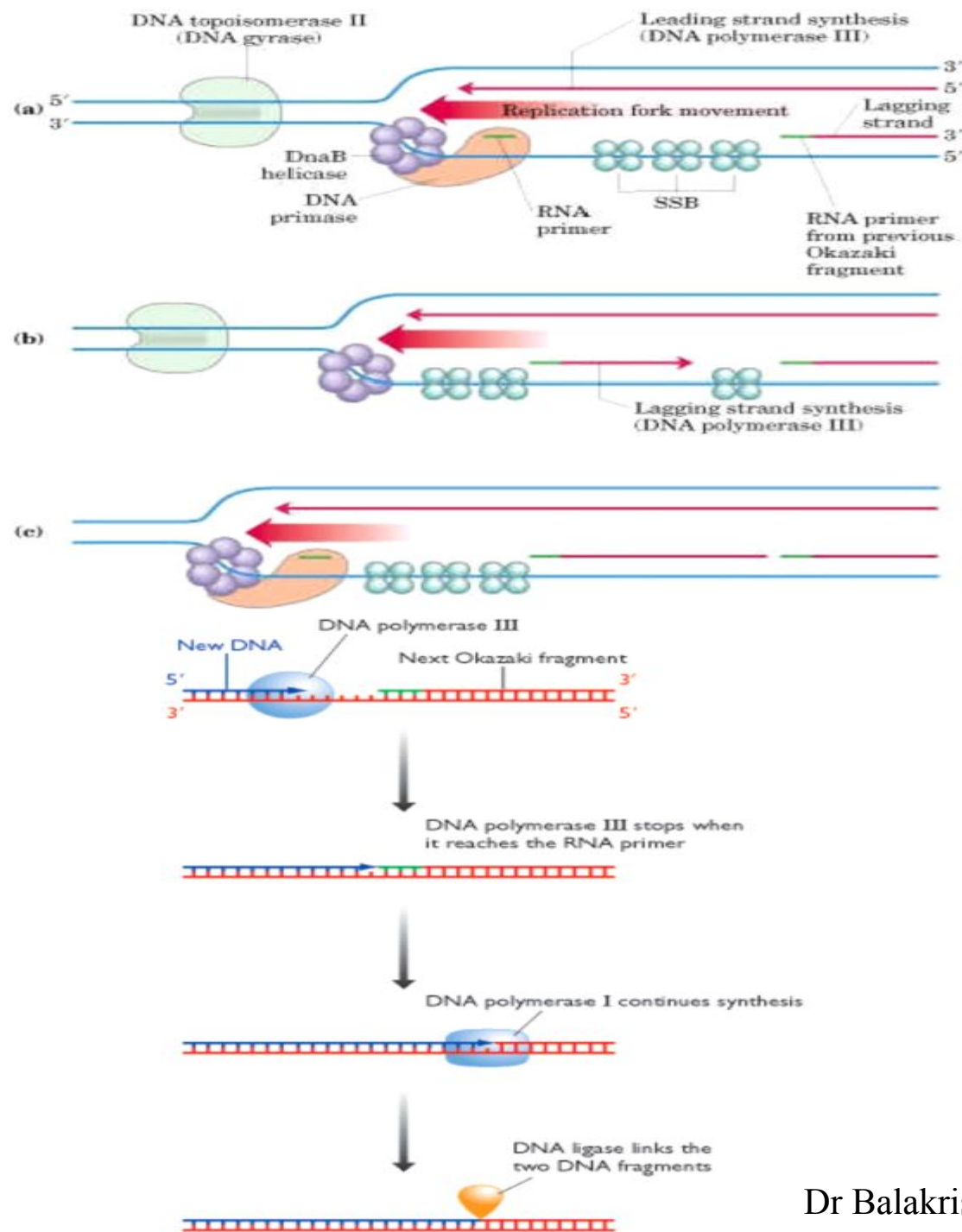




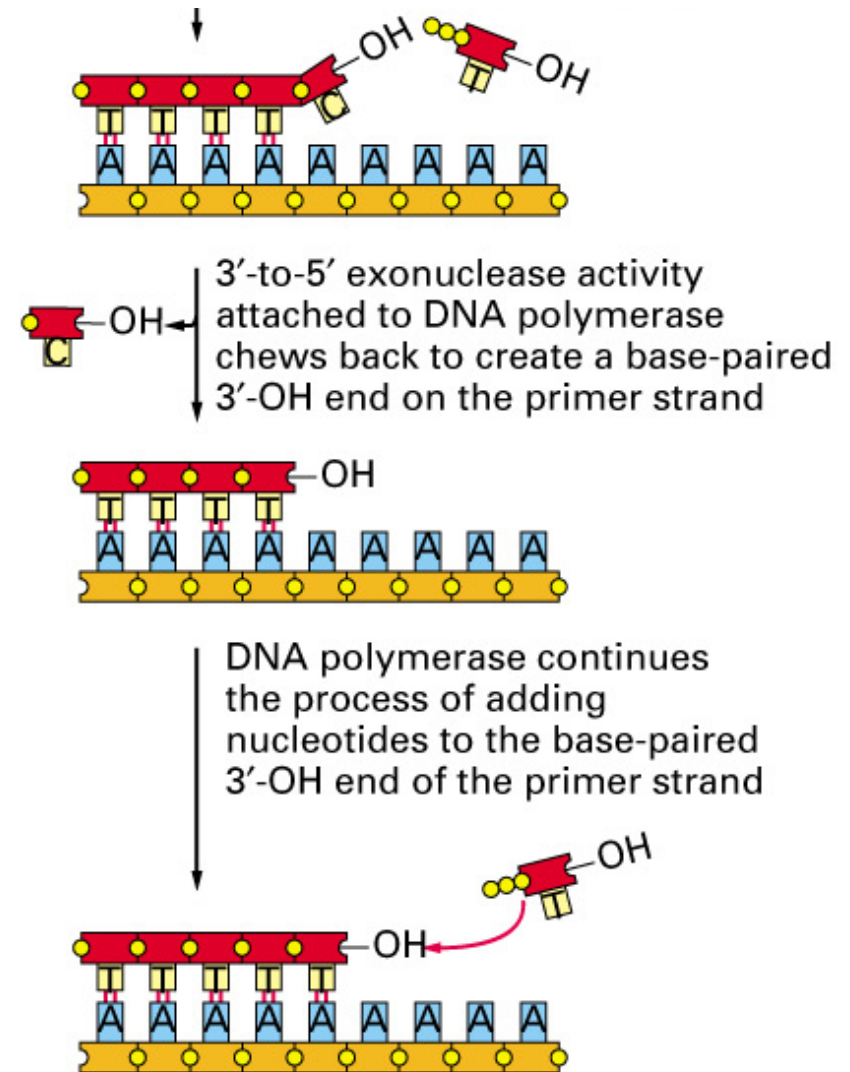
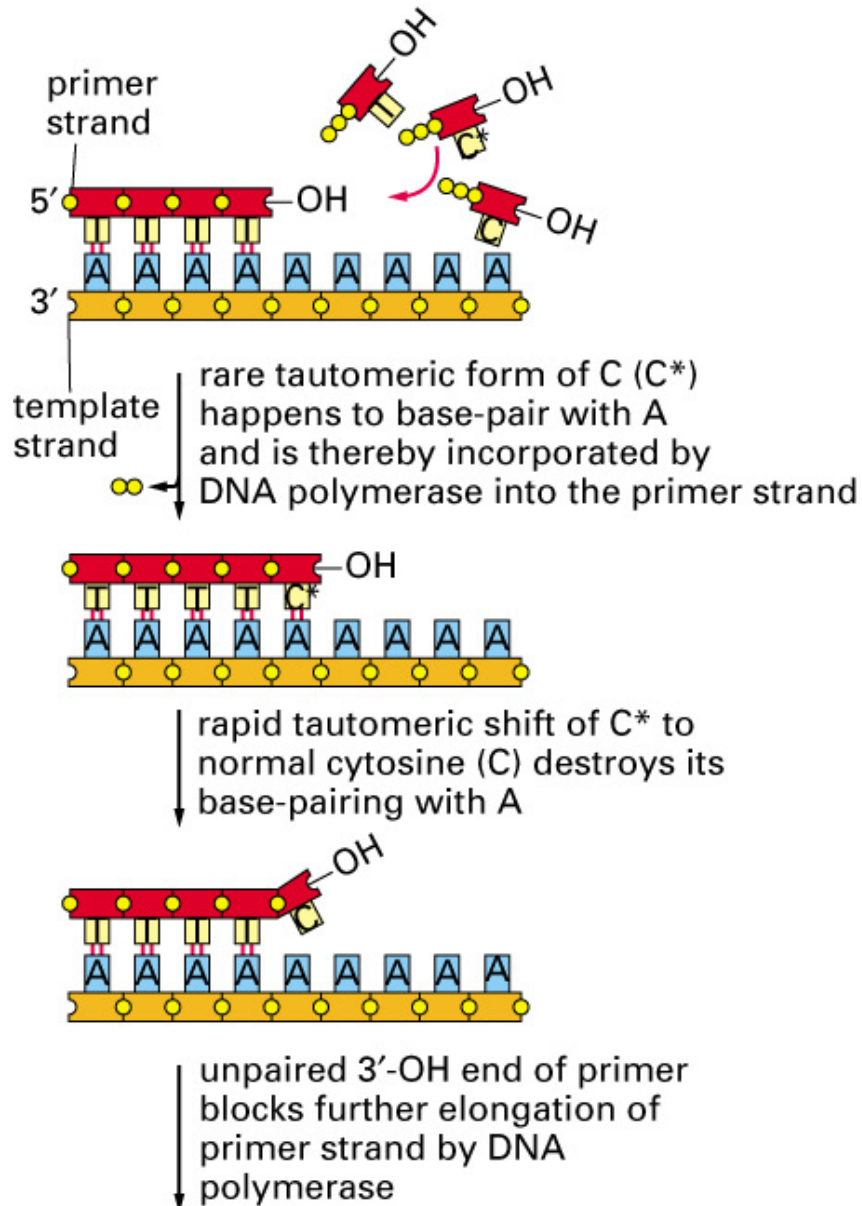
*Topoisomerase IV - Decatenates replicated DNA

Steps of DNA Replication:

6. Once all of the bases are matched up (A with T, C with G), an enzyme called exonuclease strips away the primer(s). (pol I; pol β (RNAse H))
7. The gaps where the primer(s) were are then filled by yet more complementary nucleotides. (pol I; pol β)
8. The new strand is proofread to make sure there are no mistakes in the new DNA sequence. (pol II; pol ϵ)
9. Finally, an enzyme called DNA ligase seals up the sequence of DNA into two continuous double strands.
10. The result of DNA replication is two DNA molecules consisting of one new and one old chain of nucleotides.
 - This is why DNA replication is described as semi-conservative, half of the chain is part of the original DNA molecule, half is brand new.
11. Following replication the new DNA automatically winds up into a double helix.



DNA Proofreading



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DNA Ligase

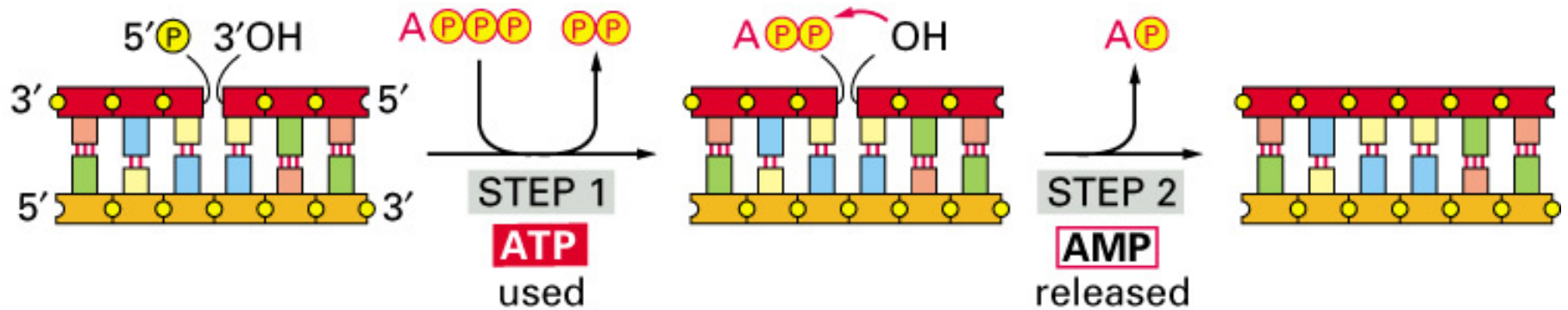


Figure 5–14. Molecular Biology of the Cell, 4th Edition.

Steps and Proteins Involved in DNA Replication

Step in Replication	Prokaryotic Cells	Eukaryotic Cells (Nuclei)
Origin of replication (ori)	One ori site per chromosome	Multiple ori sites per chromosome
Unwinding of DNA double helix	Helicase	Helicase
Stabilization of unwound template strands	Single-stranded DNA-binding protein (SSB)	Single-stranded DNA-binding protein (SSB)
Synthesis of RNA primers	Primase	Primase
Synthesis of DNA Leading strand Lagging strand (Okazaki fragments)	DNA polymerase III DNA polymerase III	DNA polymerases $\alpha + \delta$ DNA polymerases $\alpha + \delta$
Removal of RNA primers	DNA polymerase I (5'→3' exonuclease)	RNase H (5'→3' exonuclease)
Replacement of RNA with DNA	DNA polymerase I	DNA polymerase δ
Joining of Okazaki fragments	DNA ligase	DNA ligase
Removal of positive supercoils ahead of advancing replication forks	DNA topoisomerase II (DNA gyrase)	DNA topoisomerase II
Synthesis of telomeres	Not required	Telomerase

Summary of DNA Replication

Steps of DNA replication are as follows:

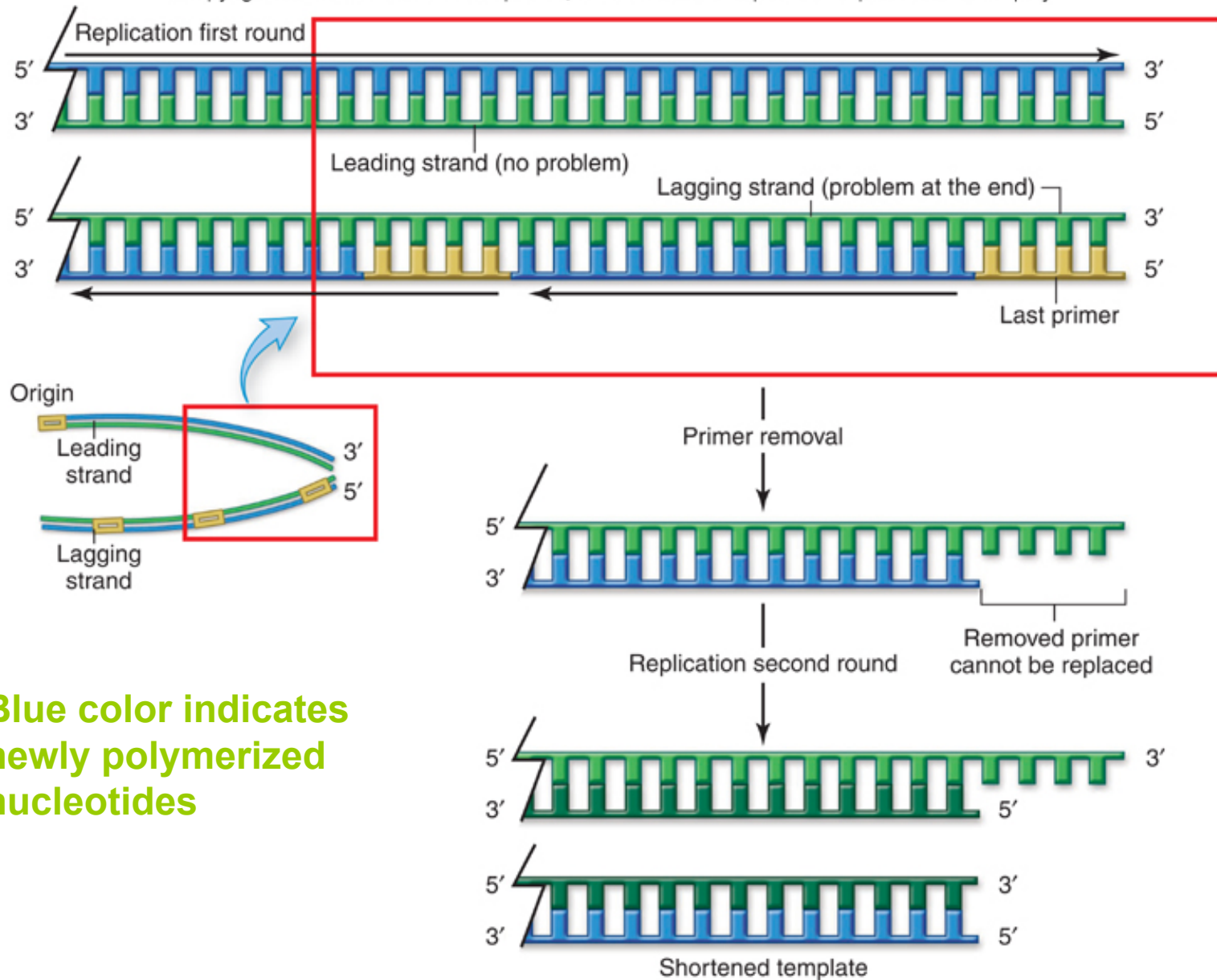
- Identification of site of the origin of replication (ori)
- Unwinding of parental DNA (dsDNA → ssDNA)
- Formation of **replication fork**
- Synthesis of RNA **Primer**, complementary to DNA template, the enzyme required is **primase**
- Leading strand is synthesised in the 5' to 3' direction by the enzyme **DNA polymerase**
- Lagging strand is synthesised as **Okazaki** fragments.
- RNA pieces are removed, when polymerisation is complete
- The gaps are filled by deoxynucleotides and the pieces are joined by the nick sealing enzyme **DNA ligase**, which requires energy from ATP.

Telomeres problem in Eukaryotic DNA Replication

- Difficult to synthesizing the ends of the chromosomes
 - Lack of a primer.
 - DNA polymerase cannot complete synthesis of the 5' end of each strand.
- With each round of DNA replication, the linear eukaryotic chromosome becomes so short
 - Those chromosomes cannot function properly and the cells die.
- **Telomeres** – repeated DNA sequence on the ends of eukaryotic chromosomes
 - Produced by **telomerase**

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Synthesis of Telomeres in Eukaryotes

- **Telomerase;**
 - Contains an **RNA region** - used as a template, as well as;
 - Reverse transcriptase activity (hTRT) - so a DNA primer can be produced.
 - Normally present only in embryonic cells, germ (reproductive) cells, and stem cells, but not in somatic cells.
 - Cancer cells often have relatively high levels of telomerase, preventing the telomeres from becoming shortened and contributing to the **immortality of malignant cells**.

Telomerase Structure

Reverse transcriptase with RNA template to bind to DNA strands

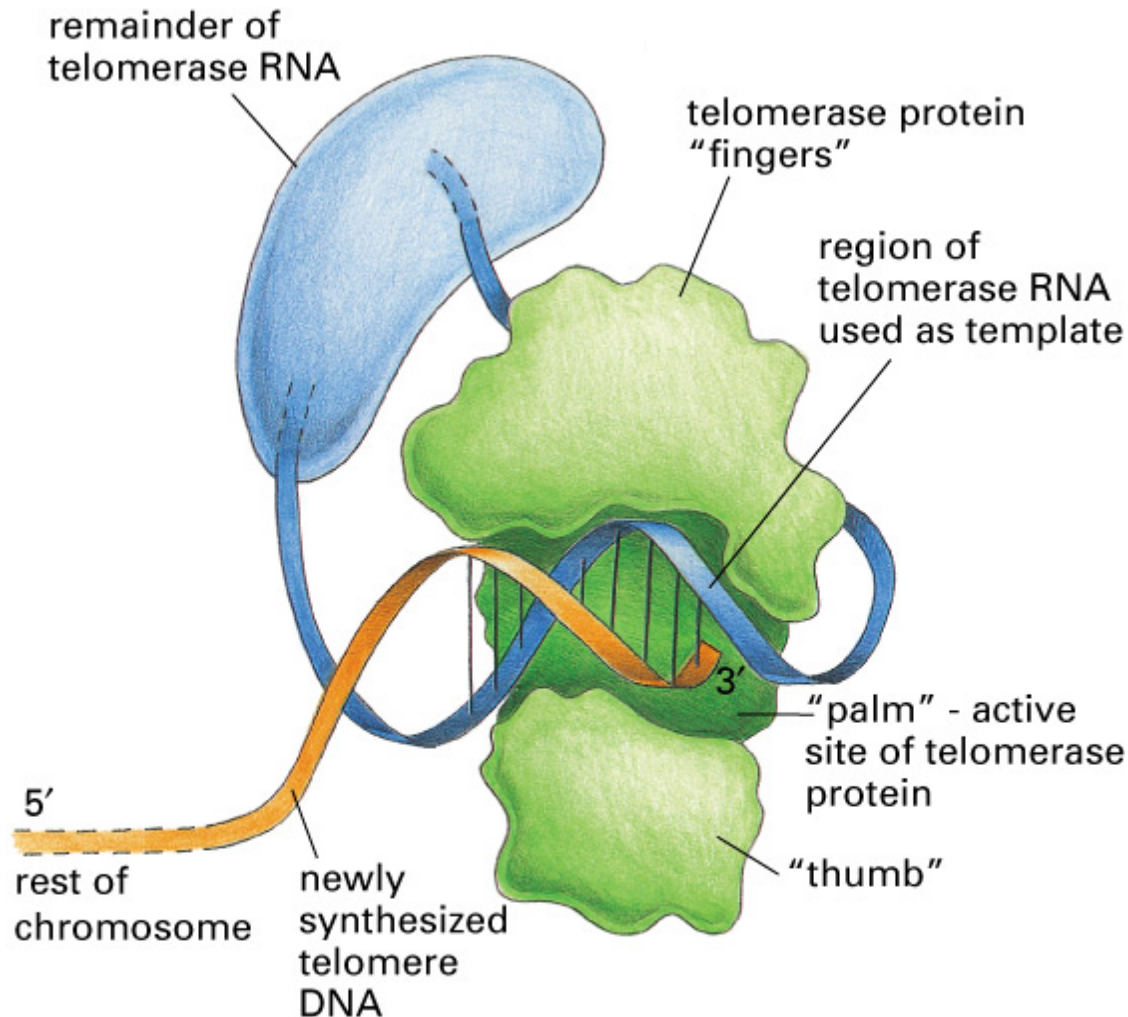
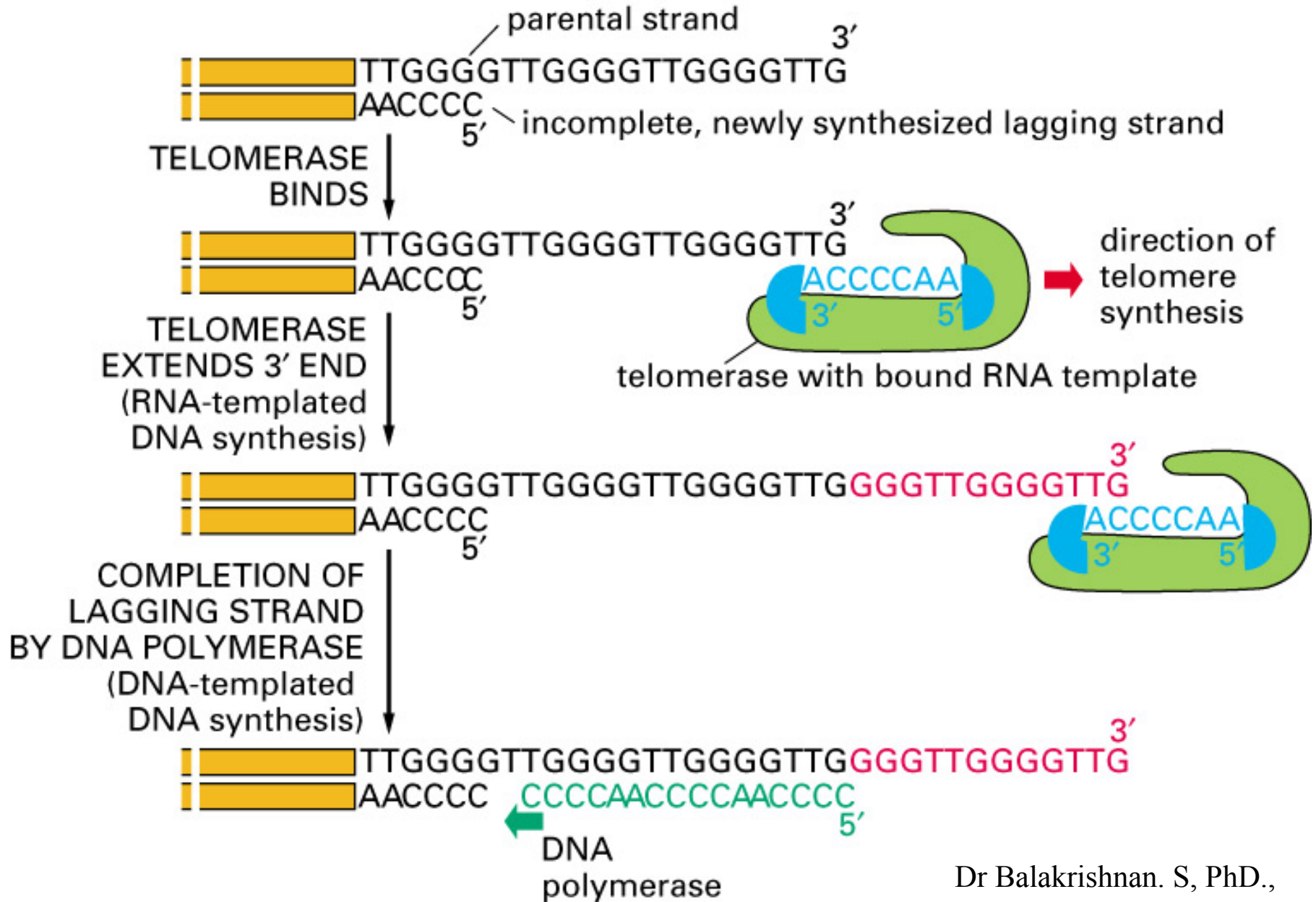


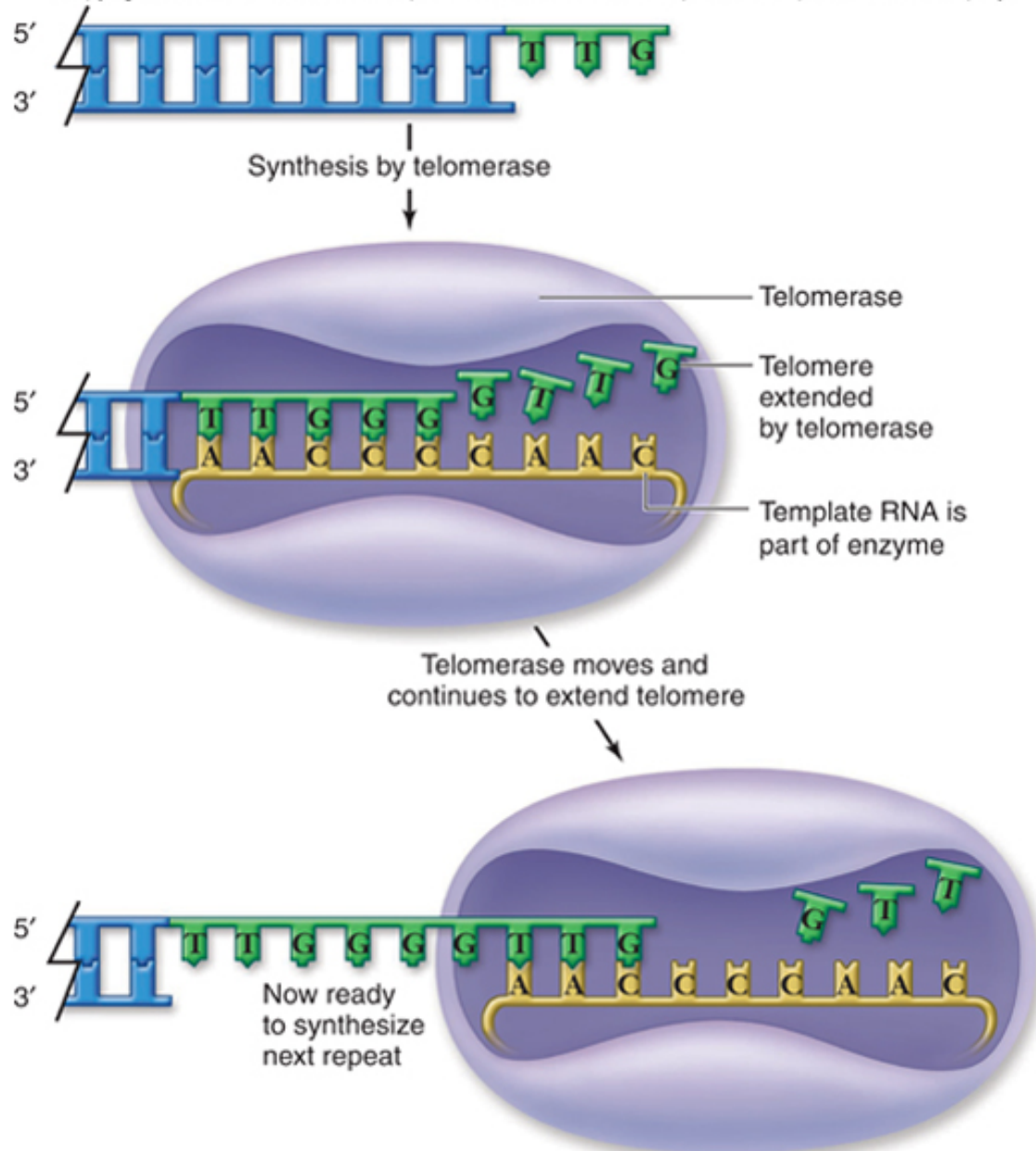
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Telomerase and its function



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Differences between replication in prokaryotes and eukaryotes

Prokaryotic DNA Replication:	Eukaryotic DNA Replication:
1. It occurs inside the cytoplasm.	1. It occurs inside the nucleus.
2. There is single origin of replication.	2. Origin of replications are numerous.
3. DNA polymerase III carries out both initiation and elongation.	3. Initiation is carried out by DNA polymerase α while elongation by DNA polymerase δ and ϵ .
4. DNA repair and gap filling are done by DNA polymerase I.	4. The same are performed by DNA polymerase β .
5. RNA primer is removed by DNA polymerase I.	5. RNA primer is removed by DNA polymerase β .
6. Okazaki fragments are large, 1000-2000 nucleotides long.	6. Okazaki fragments are short, 100-200 nucleotides long.
7. Replication is very rapid, some 2000 bp per second.	7. Replication is slow, some 100 nucleotides per second.
8. DNA gyrase is needed.	8. DNA gyrase is not needed.
9. Have no ends to synthesize	9. Having distinct process for replicating the telomeres at the ends of their chromosomes
10. Occurs almost continuously	10. Occurs only during the S-phase of the cell cycle

Inhibitors of DNA replication

- **Dauromycin & Adriamycin**: interfering with passage of both DNA and RNA polymerases, thus inhibit both replication and transcription, antibiotic.
- **Actinomycin D**: intercalating agent, inhibit both replication and transcription (antibiotic, anticancer agent).
- **Ethidium bromide & proflavin**: intercalating agent, inhibit both replication and transcription.
- **Novobiocin & oxolinic acid**: Prokaryotic DNA gyrases inhibitor, antibiotic.
- **Aphidicolin**: inhibit DNA pol α , δ and ϵ .
- **Rifamycin**: inhibit RNA pol, so RNA primer will not available. Inhibit RT also. (antibiotic, antitumor agent)

Quinolones and DNA Gyrase

- Quinolones and fluoroquinolones inhibit DNA gyrase (prokaryotic topoisomerase II), preventing DNA replication and transcription.
- These drugs, which are most active against aerobic gram-negative bacteria, include:
 - ✧ Levofloxacin
 - ✧ Ciprofloxacin
 - ✧ Moxifloxacin

Nucleotide analogues and HIV treatment

- AZT (3'-azido-2',3'-dideoxythymidine) or structurally related compounds (such as ddC, and ddI) – used in the chemotherapeutic treatment of HIV.
 - Once enter cells □ converted to the triphosphate derivative and used as a substrate for the viral reverse transcriptase.
 - The replacement or missing of a normal hydroxyl group at the 3' position of the deoxyribose prevents further replication by effectively causing chain termination.

Thank you...



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